

Home work problemsSOLUTIONS

- ① For $u(x, y)$, $0 \leq x < \infty$, the cosine Fourier transform with respect to x is:

$$U_c(p, y) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} u(x, y) \cos px \, dx.$$

$$\begin{aligned} \mathcal{F}_c[u_{yy}] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\partial^2 u}{\partial y^2} \cos px \, dx \\ &= \frac{\partial^2}{\partial y^2} \sqrt{\frac{2}{\pi}} \int_0^{\infty} u(x, y) \cos px \, dx \end{aligned} \quad \left. \begin{array}{l} \text{Integration over } x \text{ and differentiation} \\ \text{with respect to } y \\ \text{are interchangeable} \end{array} \right\}$$

$$\Rightarrow \mathcal{F}_c[u_{yy}] = \frac{\partial^2 U_c(p, y)}{\partial y^2}.$$

$$\mathcal{F}_c[u_{xx}] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\partial^2 u}{\partial x^2} \cos px \, dx \quad \left. \begin{array}{l} \text{Integrating by parts} \end{array} \right\}$$

$$= \sqrt{\frac{2}{\pi}} \left. \frac{\partial u}{\partial x} \cos px \right|_0^{\infty} - \sqrt{\frac{2}{\pi}} \int_0^{\infty} (-p \sin px) \frac{\partial u}{\partial x} \, dx$$

The contribution of the upper limit vanishes since $\frac{\partial u}{\partial x} \rightarrow 0$ as $x \rightarrow \infty$

$$= -\sqrt{\frac{2}{\pi}} u_x(0, y) + \underbrace{\sqrt{\frac{2}{\pi}} p u \sin px \Big|_0^{\infty}}_{\substack{\parallel \\ (u \xrightarrow{x \rightarrow \infty} 0)}} - p^2 \underbrace{\sqrt{\frac{2}{\pi}} \int_0^{\infty} \cos px \, u \, dx}_{U_c(p, y)}$$

Therefore:

$$\mathcal{F}_c[u_{xx}] = -\sqrt{\frac{2}{\pi}} u_x(0, y) - p^2 U_c(p, y).$$

(2) $u_{xx} + u_{yy} = 0, \quad 0 \leq x < \infty, \quad 0 \leq y < \infty,$

boundary conditions: $u_x(0, y) = 0$

$$u(x, 0) = \begin{cases} T, & 0 \leq x \leq a \\ 0, & a < x < \infty \end{cases}$$

(a) Applying the cosine Fourier transform to $u_{xx} + u_{yy} = 0$, we have:

$$-\underbrace{\sqrt{\frac{2}{\pi}} u_x(0, y)}_0 - p^2 U_c + \frac{\partial^2 U_c}{\partial y^2} = 0$$

$$\Rightarrow \frac{\partial^2 U_c}{\partial y^2} - p^2 U_c = 0.$$

The general solution of this equation is:

$$U_c(p, y) = A(p) e^{-py} + B(p) e^{py}.$$

We require that the solution does not diverge at $y \rightarrow \infty$, hence, we set $B(p) = 0$.

$$\Rightarrow U_c(p, y) = A(p) e^{-py}.$$

Applying the boundary condition at $y = 0$:

$$U_c(p, 0) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} u(x, 0) \cos px \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^a T \cos px \, dx$$

$$= \sqrt{\frac{2}{\pi}} T \left. \frac{\sin px}{p} \right|_0^a = T \sqrt{\frac{2}{\pi}} \frac{\sin pa}{p}.$$

Hence, $A(p) = T \sqrt{\frac{2}{\pi}} \frac{\sin pa}{p},$ and

$$U_c(p, y) = T \sqrt{\frac{2}{\pi}} \frac{\sin pa}{p} e^{-py}.$$

(b) To find $u(x, y)$ we perform the inverse cosine Fourier transform, (3)

$$u(x, y) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} U_c(p, y) \cos px \, dp$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} T \sqrt{\frac{2}{\pi}} \frac{\sin pa}{p} e^{-py} \cos px \, dp$$

$$= \frac{2T}{\pi} \int_0^{\infty} \frac{\sin pa \cos px}{p} e^{-py} \, dp \left\{ \begin{array}{l} \sin pa \cos px \\ = \frac{1}{2} [\sin(a+x)p \\ + \sin(a-x)p] \end{array} \right.$$

$$= \frac{T}{\pi} \int_0^{\infty} \frac{\sin(a+x)p + \sin(a-x)p}{p} e^{-py} \, dp$$

$$= \frac{T}{\pi} \left[\arctan \frac{a+x}{y} + \arctan \frac{a-x}{y} \right] \left\{ \begin{array}{l} \text{Using} \\ \int_0^{\infty} \frac{\sin ap}{p} e^{-\beta p} \, dp \\ = \arctan \frac{\alpha}{\beta} \end{array} \right.$$

Note that this answer is identical to that of Example 1(c).

The solution obtained is even, $u(-x, y) = u(x, y)$, and solving for $-\infty < x < \infty$ with

$u(x, 0) = T$ for $-a \leq x \leq a$ in Example 1, is equivalent to the problem solved in this question.

The "evenness" of the solution is guaranteed by the condition $u_x(0, y) = 0$ and even character of the cosine function in the cosine Fourier transform.

③ Heat equation: $u_t - u_{xx} = 0$, $0 \leq x < \infty$ (4)
 (here $K=1$), initial condition $u(x, 0) = 0$,
 boundary condition $u(0, t) = T$.

(a) The sine Fourier transform:

$$U_s(p, t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} u(x, t) \sin px \, dx.$$

$$\begin{aligned} \mathcal{F}_s[u_t] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\partial u}{\partial t} \sin px \, dx \\ &= \frac{\partial}{\partial t} \sqrt{\frac{2}{\pi}} \int_0^{\infty} u(x, t) \sin px \, dx \end{aligned} \quad \left. \begin{array}{l} \text{Integration over } x \text{ and} \\ \text{differentiation with} \\ \text{respect to } t \text{ are} \\ \text{interchangeable.} \end{array} \right\}$$

$$\Rightarrow \mathcal{F}_s[u_t] = \frac{\partial U_s(p, t)}{\partial t}$$

$$\begin{aligned} \mathcal{F}_s[u_{xx}] &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \frac{\partial^2 u}{\partial x^2} \sin px \, dx \quad \left. \begin{array}{l} \text{Integrating by parts} \end{array} \right\} \\ &= \underbrace{\sqrt{\frac{2}{\pi}} \frac{\partial u}{\partial x} \sin px \Big|_0^{\infty}}_{\text{Contribution of upper limit is zero, as } u \xrightarrow{x \rightarrow \infty} 0} - \sqrt{\frac{2}{\pi}} \int_0^{\infty} p \cos px \frac{\partial u}{\partial x} \, dx \end{aligned}$$

Contribution of upper limit is zero, as $u \xrightarrow{x \rightarrow \infty} 0$.

since $\frac{\partial u}{\partial x} \xrightarrow{x \rightarrow \infty} 0$

$$= -\sqrt{\frac{2}{\pi}} p \cos px \, u \Big|_0^{\infty} - \sqrt{\frac{2}{\pi}} p^2 \int_0^{\infty} \sin px \, u(x, t) \, dx$$

$$\Rightarrow \mathcal{F}_s[u_{xx}] = \sqrt{\frac{2}{\pi}} u(0, t) p - p^2 U_s(p, t).$$

(b) Applying the sine Fourier transform to the heat equation:

$$\frac{\partial U_s}{\partial t} - \sqrt{\frac{2}{\pi}} u(0, t) p + p^2 U_s = 0$$

Using the boundary condition:

$$\frac{\partial U_s}{\partial t} + p^2 U_s - \sqrt{\frac{2}{\pi}} T p = 0.$$

(c) Let us re-write the above equation:

(5)

$$\frac{\partial U_s}{\partial t} + p^2 U_s = \sqrt{\frac{2}{\pi}} T p. \quad (*)$$

This is a linear 1st-order inhomogeneous differential equation in variable t (p being a parameter). Solving the homogeneous equation first,

$$\frac{\partial U}{\partial t} + p^2 U_s = 0,$$

$$\text{we obtain: } U_s = A(p) e^{-p^2 t},$$

where $A(p)$ is an arbitrary function of p (constant with respect to t). To solve the inhomogeneous equation, replace $A(p)$ with a function of t (and p): $A(p) \rightarrow A(p, t)$

$$U_s = A(p, t) e^{-p^2 t}, \quad (1)$$

and substitute into (*):

$$\frac{\partial A}{\partial t} e^{-p^2 t} + A(-p^2) e^{-p^2 t} + p^2 A e^{-p^2 t} = \sqrt{\frac{2}{\pi}} T p$$

$$\frac{\partial A}{\partial t} = \sqrt{\frac{2}{\pi}} T p e^{p^2 t}$$

$$A = \int \sqrt{\frac{2}{\pi}} T p e^{p^2 t} dt$$

$$A(p, t) = \sqrt{\frac{2}{\pi}} T \frac{1}{p} e^{p^2 t} + C(p) \quad (2)$$

↑
arbitrary "constant"
(function of p)

$C(p)$ can be determined from the initial condition: $u(x, 0) = 0$.

$$U_s(p, 0) = \sqrt{\frac{2}{\pi}} \int_0^\infty u(x, 0) \sin p x dx = 0$$

Setting $t = 0$ in (1) and using $A(p, t)$ from (2),

$$\sqrt{\frac{2}{\pi}} \frac{T}{p} + C(p) = 0$$

$$\Rightarrow C(p) = -\sqrt{\frac{2}{\pi}} \frac{T}{p}$$

Substituting this into (2) and then into (1), we have:

$$A(p, t) = \sqrt{\frac{2}{\pi}} \frac{T}{p} e^{p^2 t} - \sqrt{\frac{2}{\pi}} \frac{T}{p}$$

$$= \sqrt{\frac{2}{\pi}} T \frac{e^{p^2 t} - 1}{p}$$

$$U_s(p, t) = \sqrt{\frac{2}{\pi}} T \frac{e^{p^2 t} - 1}{p} e^{-p^2 t}$$

$$U_s(p, t) = \sqrt{\frac{2}{\pi}} T \frac{1 - e^{-p^2 t}}{p} \quad (3)$$

The function $u(x, t)$ is found by performing the inverse sine Fourier transform:

$$u(x, t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} U_s(p, t) \sin px \, dp,$$

and substituting (3), we obtain:

$$u(x, t) = \frac{2T}{\pi} \int_0^{\infty} \frac{1 - e^{-p^2 t}}{p} \sin px \, dp.$$

The remaining integral can in fact be expressed in terms of the error function $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds$, see end of next problem. This can also be done using two table integrals: $\int_0^{\infty} \frac{\sin as}{s} ds = \frac{\pi}{2} \quad (a > 0)$ and $\int_0^{\infty} e^{-\beta s^2} \frac{\sin as}{s} ds = \frac{\pi}{2} \text{erf}\left(\frac{a}{2\sqrt{\beta}}\right)$, which gives:

$$u(x, t) = T \left[1 - \text{erf}\left(\frac{x}{2\sqrt{t}}\right) \right].$$

④ Heat equation : $u_t - u_{xx} = 0$ ($K=1$) (7)

with initial condition and boundary condition

$$u(x, 0) = 0, \quad u(0, t) = f(t).$$

Applying the sine Fourier transform to the heat equation, we have:

$$\frac{\partial U_s}{\partial t} - \sqrt{\frac{2}{\pi}} u(0, t) p + p^2 U_s = 0.$$

(see question 3, part (a)).

$$\frac{\partial U_s}{\partial t} + p^2 U_s = \sqrt{\frac{2}{\pi}} f(t) p. \quad (1)$$

This is a 1st-order linear inhomogeneous equation. Solving the homogeneous equation first,

$$\frac{\partial U_s}{\partial t} + p^2 U_s = 0$$

$$U_s = A(p) e^{-p^2 t}.$$

To solve the inhomogeneous equation, replace $A(p)$ with a function of t (and p):

$$U_s(p, t) = A(p, t) e^{-p^2 t},$$

and substitute into (1):

$$\frac{\partial A}{\partial t} e^{-p^2 t} + \cancel{A(-p^2) e^{-p^2 t}} + \cancel{p^2 A e^{-p^2 t}} = \sqrt{\frac{2}{\pi}} f(t) p$$

$$\frac{\partial A}{\partial t} = \sqrt{\frac{2}{\pi}} p f(t) e^{p^2 t} \quad (2')$$

$$A(p, t) = \int \sqrt{\frac{2}{\pi}} p f(t) e^{p^2 t} dt + \text{const} \quad (2)$$

The initial condition, $u(x, 0) = 0$, requires $U_s(p, 0) = 0$,

which in turn means $A(p, 0) = 0$. (8)

This can be fulfilled by replacing the indefinite integral in (2) by a definite integral with the limits 0 and t , and setting $\text{const} = 0$:

$$A(p, t) = \sqrt{\frac{2}{\pi}} p \int_0^t e^{p^2 \tau} f(\tau) d\tau. \quad (3)$$

[Here we replaced the integration variable by τ , to avoid confusion with the upper limit t . Obviously, (3) satisfies (2') and $A(p, 0) = 0$, as required.]

Hence:
$$U_s(p, t) = \sqrt{\frac{2}{\pi}} p e^{-p^2 t} \int_0^t e^{p^2 \tau} f(\tau) d\tau.$$

To find $u(x, t)$, substitute $U_s(p, t)$ in the inverse sine Fourier transform equation,

$$u(x, t) = \sqrt{\frac{2}{\pi}} \int_0^\infty U_s(p, t) \sin px dp$$

$$\begin{aligned} u(x, t) &= \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{\frac{2}{\pi}} p e^{-p^2 t} \int_0^t e^{p^2 \tau} f(\tau) d\tau \sin px dp \\ &= \frac{2}{\pi} \int_0^\infty \int_0^t e^{-p^2(t-\tau)} p \sin px f(\tau) d\tau dp \end{aligned}$$

Changing the order of integration:

$$u(x, t) = \frac{2}{\pi} \int_0^t f(\tau) d\tau \int_0^\infty e^{-p^2(t-\tau)} p \sin px dp$$

and using
$$\int_0^\infty e^{-\alpha p^2} p \sin \beta p dp = \sqrt{\pi} e^{-\frac{\beta^2}{4\alpha}} \frac{\beta}{4\alpha^{3/2}},$$

with $\alpha = t - \tau$, $\beta = x$, to integrate over p , we obtain:

$$u(x,t) = \frac{2}{\pi} \int_0^t f(\tau) d\tau \frac{1}{\sqrt{\pi}} e^{-\frac{x^2}{4(t-\tau)}} \frac{x}{4(t-\tau)^{3/2}}$$

$$u(x,t) = \frac{2}{\sqrt{\pi}} \int_0^t f(\tau) e^{-\frac{x^2}{4(t-\tau)}} \frac{x}{4(t-\tau)^{3/2}} d\tau.$$

Let us introduce new variable, $s = \frac{x}{2\sqrt{t-\tau}}$

$$s^2 = \frac{x^2}{4(t-\tau)}, \quad \sqrt{t-\tau} = \frac{x}{2s}$$

$$t-\tau = \frac{x^2}{4s^2}, \quad \tau = t - \frac{x^2}{4s^2}$$

$$d\tau = \frac{x^2}{2s^3} ds$$

Limits: $\tau = 0 \quad s = \frac{x}{2\sqrt{t}}$

$\tau = t \quad s \rightarrow \infty$

Performing all the changes in the above integral:

$$u(x,t) = \frac{2}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{t}}}^{\infty} f\left(t - \frac{x^2}{4s^2}\right) e^{-s^2} \frac{x}{4s^3} \frac{x^2}{2s^3} ds$$

$$u(x,t) = \frac{2}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{t}}}^{\infty} f\left(t - \frac{x^2}{4s^2}\right) e^{-s^2} ds.$$

For $f(t) = T$ (as in problem 3), this expression gives:

$$u(x,t) = \frac{2T}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{t}}}^{\infty} e^{-s^2} ds = \frac{2T}{\sqrt{\pi}} \left[\int_0^{\infty} e^{-s^2} ds - \int_0^{\frac{x}{2\sqrt{t}}} e^{-s^2} ds \right]$$

$$= T \left[\frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-s^2} ds - \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{2\sqrt{t}}} e^{-s^2} ds \right]$$

$$= T \left[\underbrace{\text{erf}(\infty)}_{=1} - \text{erf}\left(\frac{x}{2\sqrt{t}}\right) \right] = T \left[1 - \text{erf}\left(\frac{x}{2\sqrt{t}}\right) \right]$$

This is the final answer to problem 3.